Engineering Physics Question Bank

Unit-1

Q-1) State piezoelectric effect, Explain piezoelectric method for production of ultrasonic wave.

Q-2) State and explain magnetotricion method for production of ultrasonic wave.

Q-3) State any four application of ultrasonic wave.

Q-4) Explain factor affecting architectural acoustics.

Q-5) What is dielectric polarization, explain polar and non-polar dielectric and polarization.

Q-6) A cinema hall has a volume of 7500 m^3 . It is required to have reverberation time of 1.5

sec. What should be the total absorption in the hall?

Q-7) Calculate the natural frequency of 40 mm pure iron rod. Given the

density of pure iron is 7.25 x 10^3 kg/m^3 and Young's modulus 115 x 10^9

N/m^2.

Q-8) Calculate the length of iron rod which can be used to produce ultrasonic waves of 20 KHz. Density of iron is 7.23 X 10^3 kg/m^3, Young's modulus is 11.6 X 10^{10} N/m^2

Q-9) Calculate the natural frequency of the ultrasonic waves generated by a

quartz crystal having thickness of 5.5 mm. [Given Y= 8 $\times 10^{10}$ N/m^2, ρ (density)=2650 Kg/m^3]

Unit-2

Q-1) Derive an expression for diameter of nth bright and dark Newton's ring.

Q-2) Explain the construction & working of Helium Neon Laser with neat,

labeled diagram.

Q-3) Explain the construction & working of semiconductor Laser with neat, labeled diagram.

Q-4) State applications of optical fiber.

Q-5) Numerical aperture of a fiber is 0.5 and core refractive index is 1.48. Find cladding refractive index.

Q-6) Derive an expression for darkness of a thin film.

Q-7) Explain the structure of optical fiber with suitable diagram.
Q-8) Derive an expression for darkness due to reflected light for parallel thin film interference.
Q-9) Derive an expression for darkness due to reflected light for wedge shaped thin film interference.

Unit-3

Q-1) Derive Schrodinger's time independent wave equation.

Q-2) Derive Schrodinger's time dependent wave equation.

Q-3) What is Heisenberg's uncertainty principle?
Q-4) If the uncertainty in position of an electron is 4X10-10m. Calculate the uncertainty in its momentum.
Q-5) What is wave function? Explain its significance.

Unit-4

Q-1) Derive an expression for atomic packing fraction for BCC and FCC crystal structure.

Q-2) Describe construction and working of GM Counter.

Q-3) Calculate the wavelength of X-rays when a potential difference of 30 KV is applied between filament and anode.
Q-4) Calculate atomic radii for Simple cubic, Body centered, and face centered cubic structure.
Q-5) Derive the relation between lattice parameter 'a' and crystal density 'ρ'.

Copper has FCC structure and its atomic radius is 1.278 A^0. Calculate density of Cu. (Given atomic weight of Cu=63.5.)

Unit-5

Q-1) Explain B-H(hysteresis) Curve for ferromagnetic materials. Define the terms Retentivity & Coercivity.
Q-2) Explain Type-I & Type II superconductors.

Q-3) Explain Conductor, Semiconductor, and Insulator on the basis of band theory of Solids

Q-4) Define superconductivity and distinguish between Type I & Type II superconductors.

Q-5) What is nanomaterial? Explain top-down and bottom-up approach for synthesis of nanomaterial
Q-6) Define Hall effect. Derive an expression for Hall coefficient.
Q-7) Explain Diamagnetic, Paramagnetic and Ferromagnetic materials with examples.